NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

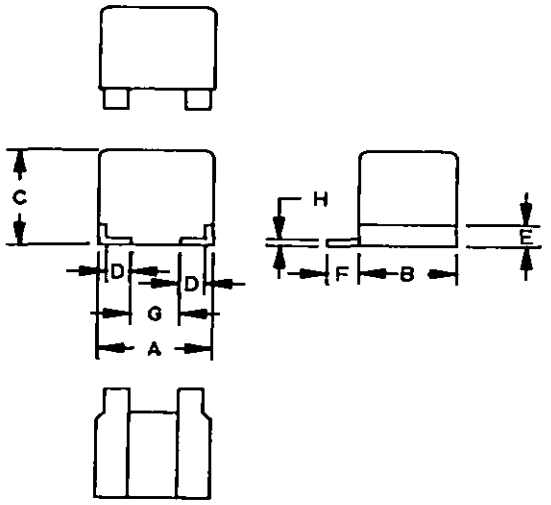
MIL-PRF-83446/11B
20 February 1985
SUPERSEDING
MIL-I-83446/11A
16 December 1980

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, SHIELDED, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

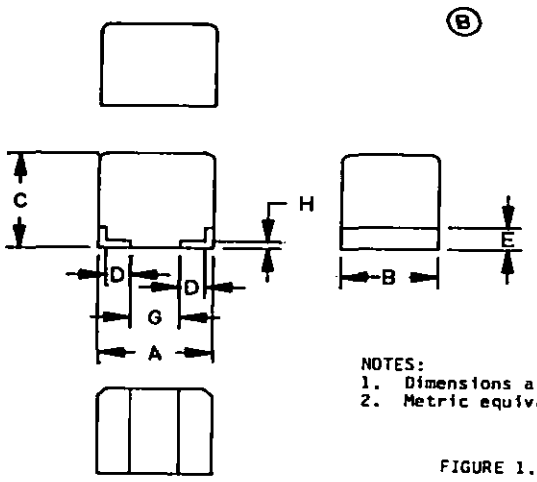
The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-83446.



	INCHES	MM
A	.155 ±.008	3.94 ±0.20
B	.125 ±.008	3.18 ±0.20
C	.130 Max	3.30 Max
D	.037 ±.003	0.94 ±0.08
E	.020 ±.005	0.51 ±0.13
F	.040 +.015 -.010	1.02 +0.38 -0.25
G	.073 Min	1.85 Min
H	.005 ±.002	0.13 ±0.05

For dash numbers 01 thru 61

(WITH TABS)



	INCHES	MM
A	.155 ±.008	3.94 ±0.20
B	.125 ±.008	3.18 ±0.20
C	.130 Max	3.30 Max
D	.037 ±.003	0.94 ±0.08
E	.020 ±.005	0.51 ±0.13
G	.073 Min	1.85 Min
H	.005 ±.002	0.13 ±0.05

For dash numbers 62 thru 122

(WITHOUT TABS)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, shielded, chip coil.

Ⓑ denotes changes

FSC 5950

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REQUIREMENTS.

Dimensions and configuration. See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range. -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude. 70,000 feet.

Dielectric withstanding voltage Method 301 of MIL-STD-202, test voltage 500 volts rms.

Barometric pressure: Method 105, test condtion C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial). See table I.

Percent coupling: 3-percent maximum, mounted with the X-Y surfaces, side by side but not touching.

Electrical characteristics (final) See table II.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number (B)	Inductance ±10%	Q (min)	Q (typ)	Test frequency 1/	Self-resonant frequency (min)	DC resistance (max)	2/ Current (max)	3/ Incremental current (max)
	µH			MHz	MHz	Ohms	mA	mA
01, 62	0.10	90	100	25	600	0.03	1000	280
02, 63	0.12	"	"	"	520	0.03	"	"
03, 64	0.15	"	"	"	490	0.03	"	"
04, 65	0.18	"	"	"	460	0.04	"	"
05, 66	0.22	"	"	"	430	0.04	"	"
06, 67	0.27	100	113	"	370	0.04	"	"
07, 68	0.33	105	120	"	310	0.05	750	200
08, 69	0.39	115	130	"	280	0.05	750	200
09, 70	0.47	115	130	"	260	0.05	750	200
10, 71	0.56	115	130	"	240	0.06	700	190
11, 72	0.68	115	130	"	200	0.06	700	190
12, 73	0.82	115	130	"	185	0.06	700	190
13, 74	1.0	115	130	"	175	0.09	650	170
14, 75	1.2	90	100	7.9	150	0.09	650	170
15, 76	1.5	100	115	"	135	0.14	600	150
16, 77	1.8	"	"	"	120	0.20	500	130
17, 78	2.2	"	"	"	105	0.30	400	100
18, 79	2.7	"	"	"	85	0.40	350	88
19, 80	3.3	"	"	"	80	0.46	330	83
20, 81	3.9	105	120	"	64	0.52	310	78
21, 82	4.7	115	130	"	56	0.54	300	75
22, 83	5.6	"	"	"	49	0.60	285	72
23, 84	6.8	"	"	"	45	0.66	270	68
24, 85	8.2	"	"	"	41	1.00	225	57
25, 86	10.0	115	130	"	39	1.20	200	50
26, 87	12.0	100	113	2.5	34	1.5	180	43
27, 88	15.0	100	113	"	30	1.8	170	41
28, 89	18.0	100	113	"	26	1.9	160	38
29, 90	22.0	105	120	"	23	2.1	150	36

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number Ⓑ	Inductance ±10%	Q (min)	Q (typ)	Test frequency 1/	Self-resonant frequency (min)	DC resistance (max)	2/ Current (max)	3/ Incremental current (max)
	µH			MHz	MHz	Ohms	mA	mA
30, 91	27.0	110	125	2.5	20	2.4	140	34
31, 92	33.0	115	130	"	18	2.7	130	32
32, 93	39.0	115	130	"	17	3.1	125	29
33, 94	47.0	115	130	"	16	3.2	125	29
34, 95	56.0	110	125	"	14	3.5	120	28
35, 96	68.0	"	"	"	12	4.0	111	26
36, 97	82.0	"	"	"	10	4.8	102	23
37, 98	100.0	"	"	"	4.4	5.7	93	20
38, 99	120.0	85	95	0.79	8.0	6.2	89	20
39, 100	150.0	"	"	"	8.0	6.3	89	20
40, 101	180.0	"	"	"	6.9	6.4	88	19
41, 102	220.0	"	"	"	6.1	7.4	82	18
42, 103	270.0	"	"	"	5.2	8.1	78	17
43, 104	330.0	100	113	"	4.6	8.8	75	16
44, 105	390.0	"	"	"	4.0	9.7	72	16
45, 106	470.0	"	"	"	3.6	10.0	69	15
46, 107	560.0	"	"	"	2.8	11.0	66	14
47, 108	680.0	"	"	"	2.3	12.0	64	13
48, 109	820.0	95	106	"	2.1	17.0	53	11
49, 110	1000.0	95	106	"	2.0	22.0	47	10
50, 111	1200.0	75	87	0.25	1.7	24.0	45	9
51, 112	1500.0	"	"	"	1.6	25.0	44	9
52, 113	1800.0	"	"	"	1.5	27.0	43	9
53, 114	2200.0	"	"	"	1.4	30.0	40	8
54, 115	2700.0	"	"	"	1.3	34.0	38	8
55, 116	3300.0	"	"	"	1.2	39.0	35	7.5
56, 117	3900.0	"	"	"	1.1	56.0	29	5.5
57, 118	4700.0	"	"	"	1.0	70.0	26	5.0
58, 119	5600.0	"	"	"	0.90	80.0	25	5.0
59, 120	6800.0	"	85	"	0.80	90.0	23	4.5
60, 121	8200.0	"	"	"	0.70	100.0	22	4.5
61, 122	10000.0	"	"	"	0.70	110.0	21	4.0

1/ Test frequency range: 25 through 150 MHz.

For electrical characteristics measurements use TF-250RX-2 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

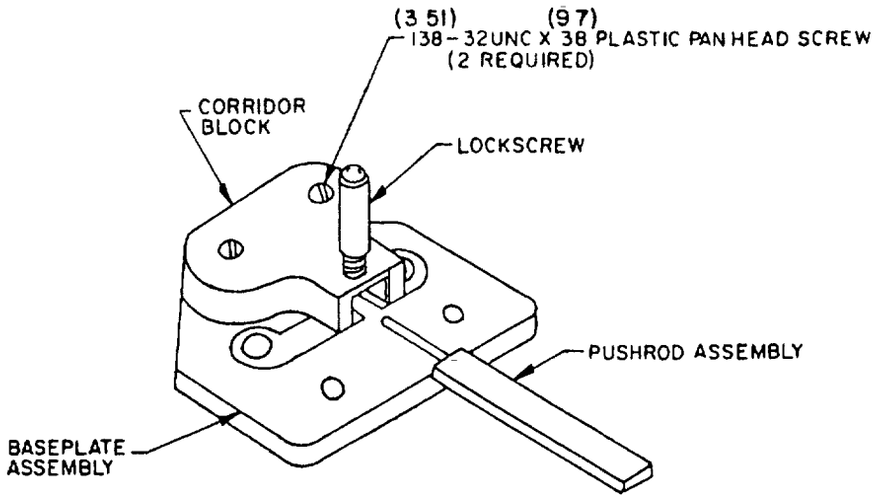
Test frequency range: 0.25 through 25 MHz.:

For electrical characteristics measurements use TF-260Q-2 test fixture, or equal (see figure 3). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

2/ Maximum current allowed not to exceed the specified temperature rise.

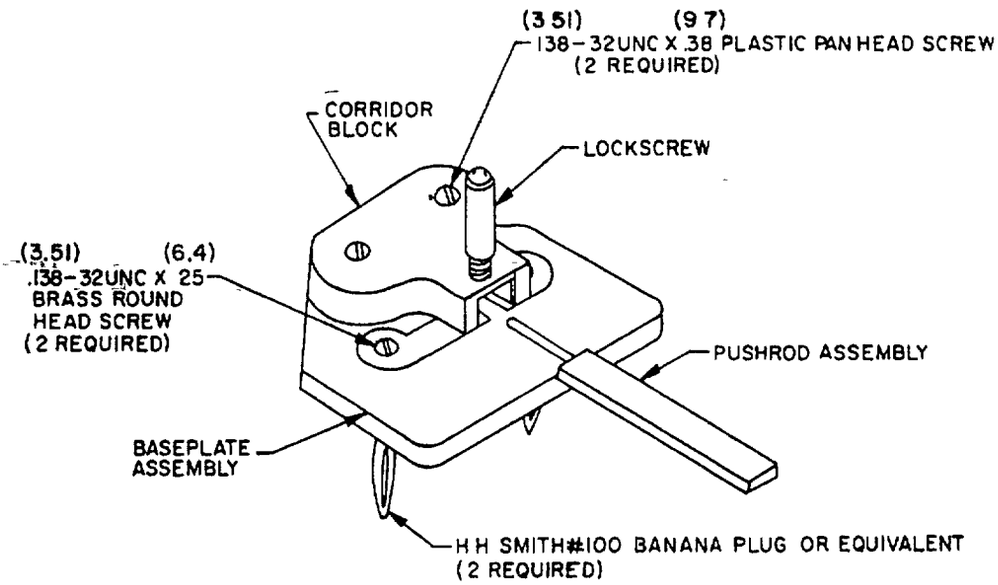
3/ Applicable to ferrite core only.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equal)



NOTES·
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3 Typical chip coil test fixture (TF-260Q-1, or equal)

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±10	±(5% +0.001 ohm)	-10	-20
Group IV	±10	±(3% +0.001 ohm)	-15	-20
Group V	±5	---	---	-20
Quality conformance inspection				
Group C Subgroup II	±10	±(5% +0.001 ohm)	-15	-20
Subgroup IV	±10	±(5% +0.001 ohm)	-15	-20

Supersession data A new part numbering system with codes for termination materials was incorporated with revision A, superseding MIL-I-83446/11, dated 1 March 1979.

M83446/11- (dash number from table I) C
supersedes M83446/11- (dash number from table I).

Examples.

M83446/11-21C supersedes M83446/11-21.
M83446/11-03C supersedes M83446/11-3.

Part number The part number shall be in the following form.

M83446/11- 04 B

04: Sequentially assigned dash numbers (see table I)

B: Termination finish (see MIL-C-83446)

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities
Navy - OS
Air Force - 11

Agent.
DLA - ES

Preparing activity.
Army - ER

(Project 5950-0633-6)

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